# Jungsoo Kim

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## Education

- **Seoul National University, Graduate School of Data Science** — Seoul, Korea
  *Master in Data Science; GPA: 4.22/4.5* — *2022.03 - 2024.08*
  ***Courses:*** *Convex Optimization, High Performance Computing, Bigdata and Knowledgement Management System, Causal Inference*
  ***Teaching Assistant:*** *Causal Inference, Machine Learning and Deep Learning II, Basic Statatistics & Mathematics Bootcamp*
- **Yonsei University, Underwood International College** — Seoul, Korea
  *B.P.A. in Science Technology and Policy (STP);* — *2017.03 - 2022.02*
  *B.B. in Quantitative Risk Management (QRM); GPA: 4.15/4.5*
  ***Courses:*** *Mathematical Statistics, Regression Analysis, Econometrics, Survival Data Analysis, Time Series Analysis, IT and Socio-Economic change, Principle of Finance Engineering*
- **National University of Singapore, Yale NUS College** — Singapore
  *Exchange Student in Mathematical and Computational Science* — *2019.09 - 2020.02*

## Publications

- **Paper: Decomposing Conformal Uncertainty: Calibration- and Instance-Driven Feature Attribution**: Accepted as a poster at NeurIPS 2026.
- **Paper: A Tale of Two Uncertainties: Global-Local Attribution for Conformal Prediction** : Spotlight in ICML Workshop on Epistemic Intelligence in Machine Learning (EIML). Sangyeon Cho*, Minyoung Cho*, **Jungsoo Kim***, Sujeong Oh, Sanghack Lee. (2026)
- **Paper: When Mass Atrocities Come to Light in the Artificial Eyes in the Sky: AI and the Use of Satellite Images for International Criminal Accountability** : Under review. **Jungsoo Kim**, Heejin Kim, Sanghack Lee. (2025)
- **Paper: Instrumental Variable Representation Learning under Confounded Covariates**: **Jungsoo Kim**, Kwonho Kim, Inwoo Hwang, Sanghack Lee. *In Neurips Workshop on CauScien: Uncovering Causality in Science.* (2025)
- **Book: Diagonal Argument - Anthology of 5th Postech SF Award-winning works**: Chapter *Will of Stochastic Ghost*, Geonryul Park, Huyeong Lee, **Jungsoo Kim**. *Eunhengnamu.* (2025) ISBN: 9791167375810

## Research and Industrial Experience

- **Ministry of Food and Drug Safety (MFDS) Project** — 2025.03-2025.12
  - **Project lead for causal inference section**: Led MFDS project on drug-effect estimation using conformal prediction (on-going)
  - **Developing methods.**: Promoted interpretability for uncertatinty quantification writing a paper as a team.
- **SK Center for Social value Enhancement Studies–Research Assistant** — 2021.08-2022.01
  - **Review and Estimation**: Reviewed beneficiaries' financial records of 10 billion won and estimated next fiscal year's budget (2022)
  - **Trend Analysis**: Researched trends of Korean social economy, focusing on economic polarization, creating a research database
- **Mooncademy (Korean Traditional Music Education Start-up)–Product Manager** — 2020.10-2021.04
  - **Planning Website IA**: Planned website IA and page layouts for a Korean traditional music platform, collaborating with engineers and designers
  - **Crowdfunding Campaign**: Launched a Wadiz crowdfunding campaign for Gayageum lectures, raising ₩10,491,700
  - **IR Presentation**: Presented the 2021 annual plan at an investor relations meeting

## Leadership and Involvement

- **Coding Instructor for Malaysian High School Students**: (at Seoul National University Language Education Institute) Taught Python programming and basic algorithms to Malaysian high school students as part of an SNU-led outreach program. Designed interactive sessions emphasizing teamwork and oral presentation to discourage reliance on generative AI tools and foster authentic problem-solving. (2025.07-2025.08)
- **Manager and Main Speaker for STP–STS Forum**: Organized and hosted a department forum for students and faculty in Science & Technology Studies in Seoul National Univ. and Science & Technology Policy in Yonsei Univ., simultaneously researching legal controversies in medical law regarding tattooing. (2022.05)
- **Project Management Director at Social Innovation Creators' Academy**: Managed cross-team project schedules and deliverables; coordinated milestones across multiple teams while developing a semester-long project roadmap and providing progress oversight. (2019.03-2020.06)
- **Yonsei Business College Drama Club (Yonsang-geuk-u-hoi)**: (Led production planning and mediated creative discussions to align diverse viewpoints and deliver staged performances participating in five productions; designed a fully remote event in response to COVID-19 constraints. (2017.03-2021.10)
- **Vice President in Major Student Council**: Liaised with faculty to address curricular and administrative issues; planned and executed major info sessions and inter-major forums regarding science-fiction. (2017.11-2018.11)

## Honors and Awards

- 2025 5th Postech SF Award (Postech); Excellence Prize – Authored a short story exploring the conflicts surrounding generative-AI–based evidence
- 2024 Journal of Korea Artificial Intelligence Association (Korea Artificial Intelligence Association); Best Paper– Submitted a paper of master's project on Instrumental Variable
- 2024 CSES Social Value Research Idea Award (Center for Social value Enhancement Studies); Excellence Prize– Proposed a study on causal structure analysis for Korean population policy
- 2024 LH COMPAS Land & City Data Competition (Korea Land & Housing Corporation); Excellence Prize – Identified medically underserved areas in Asan through geospatial analysis
- 2023 Edutech Data Analysis Hackathon (Society for EdTech); Grand Prize – Built a dropout-prediction model using survival analysis
- 2020 Public BigData Analysis Hackathon (National Information Society Agency); Excellence Prize – Developed a traffic-accident risk model with XGBoost and interpreted drivers using SHAP
- 2020 Art Data Hackathon (Arts Council Korea) – Designed a traditional musical-instrument recommendation system

## Skills and Language

- Acquired advanced knowledge of computer coding **(R, Python,C++,C)** and DBMS **(SQL, Neo4J, BigQuery, Kafka)**
- Experienced in machine learning, deep learning, and causal inference workflows, including **fine-tuning Super Resolution models** (Computer Vision; SRCNN, SRGAN, StableSR), **training Stable Diffusion with PyTorch** (Multi-modal), conducting **geospatial data analysis** using geoPandas and QGIS, and applying **causal inference methods** such as Instrumental Variable analysis and Conformal Prediction
- Fluent in both English and Korean **(TOEFL 100, OPIc (AL))**
- Comfortable with **Microsoft Word, PowerPoint, Overleaf and Excel** to utilize them at work